Instructions for the processing of PFS3

Material data

Base material			Polyoxymethylene – copolymer (POM)
Density	ρ	g/cm³	1.4
Melt temperature		°C	160 – 175
Ignition temperature		°C	320 – 340

General instructions for the processing:

Processing of PFS3 is possible using established methods for thermoplastic materials such as injection moulding and extrusion.

For the mold construction use runners as short as possible and runner cross section as large as possible.

Using injection moulding the following parameters provide a basis for further optimisation:

cylinder temperature	Zone 1	160°C
	Zone 2	170°C
	Zone 3	180°C
	nozzle	190°C
mold temperature	90-120°C	
screw speed	50 min^{-1}	
injection speed	10 cm^3/s	
melt cushion	5 mm	
injection pressure	900 bar	
hold pressure	900 bar	
time hold pressure	3 s	
back pressure	0 bar	

Temperatures above 200°C, screw speed more than 50 min^{-1}, and mold temperature above 140°C must be avoided.
Screws which develop only low shear are preferred. Low compression ratio (1 : 1.6) is recommended, maximum is 1 : 2.

Finishing:
PFS3 parts can be machined in a subsequent process.

Alle Angaben ohne Gewähr. Irrtümer und Änderungen vorbehalten. No responsibility is taken for the correctness. Errors and changings reserved.

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Safety instructions for the processing:

For common use there is no release of formaldehyde expected.
Excessively long residence times of the melt in the machine may lead to binder degradation and have to be prevented.
Avoid process temperatures above 200°C. Above 200°C gaseous decomposition products like formaldehyde can cause a health and environmental risk.

Further information on safety, health and environmental risks and on handling and storage of the polymere ingredients are given in the data sheet ‘Ultraform S2320 003 pro uncolored‘.

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